# Managing Process in Linux

## Linux Process Management

### 1. Viewing All process

```
ps aux
```

**This command shows all process running on your system .**

### explanation :

a : Show processes for all users.

u : Display the user/owner of the process.

x : Include processes not attached to a terminal (e.g., background or system services).

### Output :

```
ashishkumar@ashishkumar:~$ ps aux
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.1  0.2  23308 14288 ?        Ss   10:39   0:04 /sbin/init splash
root           2  0.0  0.0      0     0 ?        S    10:39   0:00 [kthreadd]
root           3  0.0  0.0      0     0 ?        S    10:39   0:00 [pool_workqueue_release]
root           4  0.0  0.0      0     0 ?        I<   10:39   0:00 [kworker/R-rcu_gp]
root           5  0.0  0.0      0     0 ?        I<   10:39   0:00 [kworker/R-sync_wq]
root           6  0.0  0.0      0     0 ?        I<   10:39   0:00 [kworker/R-kvfree_rcu_reclaim]
root           7  0.0  0.0      0     0 ?        I<   10:39   0:00 [kworker/R-slub_flushwq]
root           8  0.0  0.0      0     0 ?        I<   10:39   0:00 [kworker/R-netns]
root          11  0.0  0.0      0     0 ?        I<   10:39   0:00 [kworker/0:0H-kblockd]
root          12  0.0  0.0      0     0 ?        I    10:39   0:00 [kworker/u16:0-ipv6_addrconf]
root          13  0.0  0.0      0     0 ?        I<   10:39   0:00 [kworker/R-mm_percpu_wq]
root          14  0.0  0.0      0     0 ?        I    10:39   0:00 [rcu_tasks_kthread]
root          15  0.0  0.0      0     0 ?        I    10:39   0:00 [rcu_tasks_rude_kthread]
root          16  0.0  0.0      0     0 ?        I    10:39   0:00 [rcu_tasks_trace_kthread]
root          17  0.0  0.0      0     0 ?        S    10:39   0:00 [ksoftirqd/0]
root          18  0.0  0.0      0     0 ?        I    10:39   0:01 [rcu_preempt]
root          19  0.0  0.0      0     0 ?        S    10:39   0:00 [rcu_exp_par_gp_kthread_worker/0]
root          20  0.0  0.0      0     0 ?        S    10:39   0:00 [rcu_exp_gp_kthread_worker]
root          21  0.0  0.0      0     0 ?        S    10:39   0:00 [migration/0]
root          22  0.0  0.0      0     0 ?        S    10:39   0:00 [idle_inject/0]
root          23  0.0  0.0      0     0 ?        S    10:39   0:00 [cpuhp/0]
root          24  0.0  0.0      0     0 ?        S    10:39   0:00 [cpuhp/1]
root          25  0.0  0.0      0     0 ?        S    10:39   0:00 [idle_inject/1]
root          26  0.0  0.0      0     0 ?        S    10:39   0:00 [migration/1]
root          27  0.0  0.0      0     0 ?        S    10:39   0:00 [ksoftirqd/1]
root          29  0.0  0.0      0     0 ?        I<   10:39   0:00 [kworker/1:0H-events_highpri]
root          30  0.0  0.0      0     0 ?        S    10:39   0:00 [cpuhp/2]
root          31  0.0  0.0      0     0 ?        S    10:39   0:00 [idle_inject/2]
root          32  0.0  0.0      0     0 ?        S    10:39   0:00 [migration/2]
root          33  0.0  0.0      0     0 ?        S    10:39   0:00 [ksoftirqd/2]
root          35  0.0  0.0      0     0 ?        I<   10:39   0:00 [kworker/2:0H-events_highpri]
root          36  0.0  0.0      0     0 ?        S    10:39   0:00 [cpuhp/3]
```

## 2. process tree command

```
pstree -p
```

**This shows a tree-like hierarchy of all processes, starting from the init or systemd process (PID 1).**

**Output Example :**

```
systemd─┬─NetworkManager───dhclient
        ├─sshd───sshd───bash───pstree
        └─cron
```

**Shows parent-child process relationship.**

### Output :

```
ashishkumar@ashishkumar:~$ pstree -p
systemd(1)─┬─ModemManager(888)─┬─{ModemManager}(910)
           │                   ├─{ModemManager}(915)
           │                   └─{ModemManager}(921)
           ├─NetworkManager(775)─┬─{NetworkManager}(855)
           │                     ├─{NetworkManager}(861)
           │                     └─{NetworkManager}(868)
           ├─accounts-daemon(744)─┬─{accounts-daemon}(780)
           │                      ├─{accounts-daemon}(781)
           │                      └─{accounts-daemon}(789)
           ├─avahi-daemon(707)───avahi-daemon(774)
           ├─colord(1372)─┬─{colord}(1379)
           │              ├─{colord}(1380)
           │              └─{colord}(1382)
           ├─cron(751)
           ├─cups-browsed(1113)─┬─{cups-browsed}(1141)
           │                    ├─{cups-browsed}(1142)
           │                    └─{cups-browsed}(1143)
           ├─cupsd(1083)─┬─dbus(1099)
           │             ├─dbus(1100)
           │             └─dbus(1101)
           ├─dbus-daemon(709)
           ├─fwupd(4853)─┬─{fwupd}(4854)
           │             ├─{fwupd}(4855)
           │             ├─{fwupd}(4856)
           │             ├─{fwupd}(4857)
           │             └─{fwupd}(4859)
           ├─gdm3(1103)─┬─gdm-session-wor(1728)─┬─gdm-wayland-ses(1822)─┬─gnome-session-b(1830)─┬─{gnome-session-b}(1908)
           │            │                       │                       │                       ├─{gnome-session-b}(1909)
           │            │                       │                       │                       └─{gnome-session-b}(1910)
           │            │                       │                       ├─{gdm-wayland-ses}(1825)
           │            │                       │                       ├─{gdm-wayland-ses}(1826)
           │            │                       │                       └─{gdm-wayland-ses}(1829)
           │            │                       ├─{gdm-session-wor}(1729)
           │            │                       ├─{gdm-session-wor}(1730)
           │            │                       └─{gdm-session-wor}(1731)
           │            ├─{gdm3}(1105)
```

## 3. Real-Time Monitoring command :

```
top
```

**Explanation :**

top command shows a live, updating list of processes, sorted by CPU usage by default. It helps you:

Monitor CPU, memory, and process usage

Identify resource hogs

Kill or manage processes interactively

**System summary :**

```
top - 10:42:30 up  1:23,  1 user,  load average: 0.23, 0.14, 0.08
Tasks: 132 total,   1 running, 131 sleeping,   0 stopped,   0 zombie
%Cpu(s):  3.0 us,  1.0 sy,  0.0 ni, 95.7 id,  0.0 wa,  0.0 hi,  0.3 si,  0.0
st
MiB Mem :   7853.6 total,  2341.1 free,  2890.4 used,  2622.1 buff/cache
MiB Swap:   2048.0 total,  2048.0 free,     0.0 used.  4321.5 avail Mem
```

**explaination :**

| Line | Explanation |
|---|---|
| `top - 10:42:30 ...` | Current time, system uptime, number of users, load average (1, 5, 15 minutes) |
| `Tasks:` | Total number of processes and their states (running, sleeping, zombie, etc.) |
| `%Cpu(s):` | CPU usage breakdown (user, system, idle, etc.) |
| `MiB Mem:` | Total, used, free, and cached memory |
| `MiB Swap:` | Swap memory stats |

**Output :**

```
top - 11:34:46 up 55 min,  1 user,  load average: 0.44, 0.51, 0.40
Tasks: 251 total,   1 running, 249 sleeping,   1 stopped,   0 zombie
%Cpu(s):  0.3 us,  1.0 sy,  0.0 ni, 98.6 id,  0.0 wa,  0.0 hi,  0.2 si,  0.0 st
MiB Mem :   5783.9 total,   1287.7 free,   2481.2 used,   2226.7 buff/cache
MiB Swap:      0.0 total,      0.0 free,      0.0 used.   3302.7 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
   1976 ashishk+  20   0 5146556 511004 174136 S  13.0   8.6   3:31.72 gnome-shell
   3362 ashishk+  20   0  702460  59736  46876 S   2.0   1.0   0:05.23 gnome-terminal-
   3411 root      20   0       0      0      0 I   0.7   0.0   0:05.48 kworker/u20:0-events_unbound
      1 root      20   0   23308  14288   9424 S   0.0   0.2   0:04.47 systemd
      2 root      20   0       0      0      0 S   0.0   0.0   0:00.04 kthreadd
      3 root      20   0       0      0      0 S   0.0   0.0   0:00.00 pool_workqueue_release
      4 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-rcu_gp
      5 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-sync_wq
      6 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-kvfree_rcu_reclaim
      7 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-slub_flushwq
      8 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-netns
     11 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/0:0H-kblockd
     12 root      20   0       0      0      0 I   0.0   0.0   0:00.00 kworker/u16:0-ipv6_addrconf
     13 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-mm_percpu_wq
     14 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_kthread
     15 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_rude_kthread
     16 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_trace_kthread
     17 root      20   0       0      0      0 S   0.0   0.0   0:00.09 ksoftirqd/0
     18 root      20   0       0      0      0 I   0.0   0.0   0:02.45 rcu_preempt
     19 root      20   0       0      0      0 S   0.0   0.0   0:00.00 rcu_exp_par_gp_kthread_worker/0
     20 root      20   0       0      0      0 S   0.0   0.0   0:00.14 rcu_exp_gp_kthread_worker
     21 root      rt   0       0      0      0 S   0.0   0.0   0:00.09 migration/0
     22 root     -51   0       0      0      0 S   0.0   0.0   0:00.00 idle_inject/0
     23 root      20   0       0      0      0 S   0.0   0.0   0:00.00 cpuhp/0
     24 root      20   0       0      0      0 S   0.0   0.0   0:00.00 cpuhp/1
     25 root     -51   0       0      0      0 S   0.0   0.0   0:00.00 idle_inject/1
     26 root      rt   0       0      0      0 S   0.0   0.0   0:00.89 migration/1
     27 root      20   0       0      0      0 S   0.0   0.0   0:00.44 ksoftirqd/1
     29 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/1:0H-events_highpri
     30 root      20   0       0      0      0 S   0.0   0.0   0:00.00 cpuhp/2
```

## 4. Adjust Process Priority

### start a process with low priority :

```
nice -n 10 sleep 300 &
```

**explanation :**

nice: Runs a command with a modified priority (a “niceness” level).

-n 10: Sets the niceness value to +10, meaning the process will run with lower priority than the default (0).

sleep 300: The actual command being run — it just waits for 300 seconds (5 minutes) without doing anything.

&: Runs the command in the background, so your terminal remains usable.

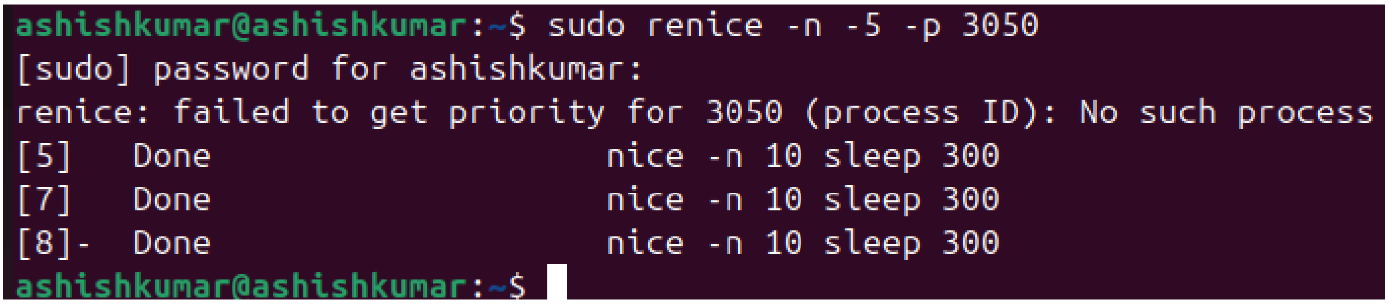
**output :**

![a3cbe25c84c8b35e570f43a41cd19c46.png](:/82225c0632c247a6913e047afc43022f

**Change priority of running process :**

```
renice -n -5 -p 3050
```

**Output :**




## 5. CPU Affinity (Bind process to CPU Core )

```
taskset -cp 3050
```

**Explanation:**

taskset: Tool to view or set CPU affinity for a process — i.e., which CPUs a process is allowed to run on.

-c: Use CPU list format (e.g., 0,1, 0-3)

-p: Operate on an existing process

3050: The PID of the process you're querying or modifying

**example output :**

```
pid 3050's current affinity list: 0-3
```

**This means the process can run on CPU cores 0, 1, 2, and 3**

## 6. I/O Scheduling priority c

## Comand :

```
ionice -c 3 -p 3050
```

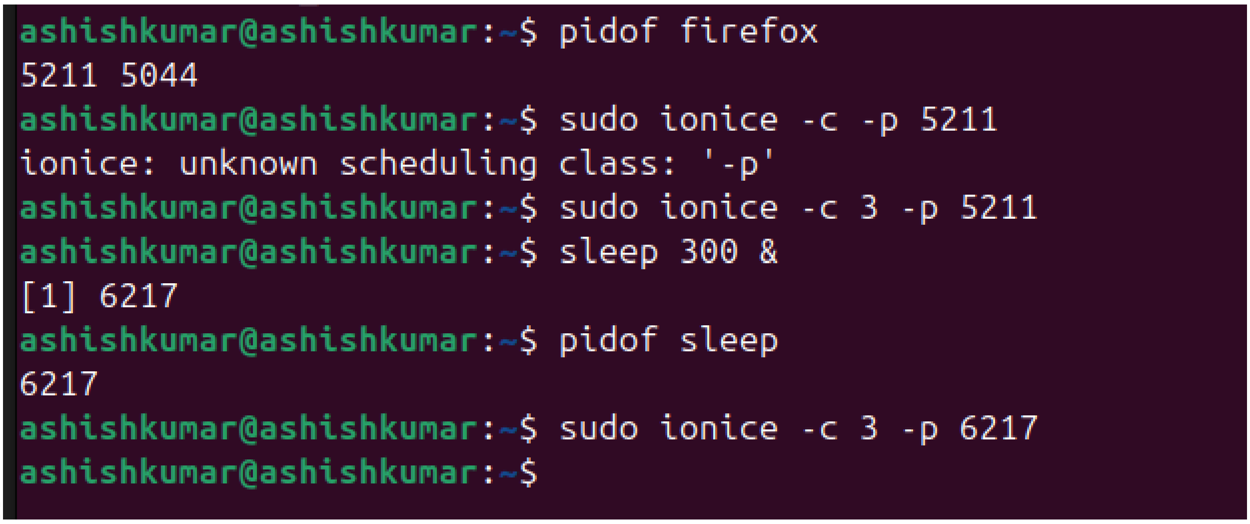
**is used to set the I/O scheduling class of a process with PID 3050 to idle class using the ionice tool on Linux.**

**Explanation : ionice: Tool to set or get the I/O scheduling class and priority of a process.**

**-c 3: Specifies the I/O scheduling class.**

**3 = Idle class. The process only gets I/O time when no other process needs it.**

**-p 3050: Specifies the process ID (PID) of the target process. In this case, PID 3050.**

**Output :** 

```
ashishkumar@ashishkumar:~$ pidof firefox
5211 5044
ashishkumar@ashishkumar:~$ sudo ionice -c -p 5211
ionice: unknown scheduling class: '-p'
ashishkumar@ashishkumar:~$ sudo ionice -c 3 -p 5211
ashishkumar@ashishkumar:~$ sleep 300 &
[1] 6217
ashishkumar@ashishkumar:~$ pidof sleep
6217
ashishkumar@ashishkumar:~$ sudo ionice -c 3 -p 6217
ashishkumar@ashishkumar:~$
```

## 7. File Descriptors Used by a process

## command :

```
lsof -p 3050 | head -5
```

**Explanation :**

lsof: Lists open files (everything is a file in Unix/Linux: files, sockets, pipes, etc.).

-p 3050: Filter results to only show those for PID 3050.

| head -5: Limit the output to the first 5 lines for readability.

**Example Output :**

```
firefox  4321 user  cwd       DIR                8,1       4096      131072
/home/user
firefox  4321 user  rtd       DIR                8,1       4096           2 /
```

```
firefox  4321 user  txt       REG                 8,1  52590496    3932161
/usr/lib/firefox/firefox
firefox  4321 user  mem       REG                 8,1    476640    3932180
/usr/lib/libgtk-3.so.0.2404.16
```

**explanation of output :**

| Column | Meaning |
|---|---|
| `COMMAND` | Name of the command or executable |
| `PID` | Process ID |
| `USER` | Username running the process |
| `FD` | File descriptor (e.g., `cwd` = current working dir, `txt` = text code) |
| `TYPE` | Type of file (e.g., `DIR`, `REG`, `CHR`, etc.) |
| `DEVICE` | Device identifier |
| `SIZE/OFF` | File size or offset |
| `NODE` | Inode number |
| `NAME` | Name of the file/path |

## 8. Trace System Calls of a process

## Command :

```
strace -p 3050
```

### Explanation :

**strace – A powerful diagnostic, debugging, and instructional tool that monitors system calls and signals used by a process.**

**-p 3050 – Tells strace to attach to the process with PID 3050 and start tracing it in real-time.**

### output:

```
ashishkumar@ashishkumar:~$ pidof firefox
5211 5044
ashishkumar@ashishkumar:~$ sudo strace -p 5044
strace: Process 5044 attached
restart_syscall(<... resuming interrupted read ...>) = 1
read(39, "\372", 1)                     = 1
newfstatat(AT_FDCWD, "/snap/firefox/6836/gnome-platform/etc/fonts/fonts.conf", {st_mode=S_IFREG|0644, st_size=2808, ...}, 0) = 0
newfstatat(AT_FDCWD, "/snap/firefox/6836/gnome-platform/etc/fonts/conf.d", {st_mode=S_IFDIR|0755, st_size=3530, ...}, 0) = 0
newfstatat(AT_FDCWD, "/snap/firefox/6836/gnome-platform/etc/fonts/conf.d/10-antialias.conf", {st_mode=S_IFREG|0644, st_size=225, ...}, 0) = 0
newfstatat(AT_FDCWD, "/snap/firefox/6836/gnome-platform/etc/fonts/conf.d/10-hinting-slight.conf", {st_mode=S_IFREG|0644, st_size=696, ...}, 0) = 0
newfstatat(AT_FDCWD, "/snap/firefox/6836/gnome-platform/etc/fonts/conf.d/10-scale-bitmap-fonts.conf", {st_mode=S_IFREG|0644, st_size=2228, ...}, 0) = 0
newfstatat(AT_FDCWD, "/snap/firefox/6836/gnome-platform/etc/fonts/conf.d/11-lcdfilter-default.conf", {st_mode=S_IFREG|0644, st_size=771, ...}, 0) = 0
newfstatat(AT_FDCWD, "/snap/firefox/6836/gnome-platform/etc/fonts/conf.d/20-unhint-small-dejavu-lgc-sans-mono.conf", {st_mode=S_IFREG|0644, st_size=874
, ...}, 0) = 0
newfstatat(AT_FDCWD, "/snap/firefox/6836/gnome-platform/etc/fonts/conf.d/20-unhint-small-dejavu-lgc-sans.conf", {st_mode=S_IFREG|0644, st_size=864, ...
}, 0) = 0
newfstatat(AT_FDCWD, "/snap/firefox/6836/gnome-platform/etc/fonts/conf.d/20-unhint-small-dejavu-lgc-serif.conf", {st_mode=S_IFREG|0644, st_size=866, ..
.}, 0) = 0
newfstatat(AT_FDCWD, "/snap/firefox/6836/gnome-platform/etc/fonts/conf.d/20-unhint-small-dejavu-sans-mono.conf", {st_mode=S_IFREG|0644, st_size=866, ..
.}, 0) = 0
newfstatat(AT_FDCWD, "/snap/firefox/6836/gnome-platform/etc/fonts/conf.d/20-unhint-small-dejavu-sans.conf", {st_mode=S_IFREG|0644, st_size=856, ...}, 0
) = 0
newfstatat(AT_FDCWD, "/snap/firefox/6836/gnome-platform/etc/fonts/conf.d/20-unhint-small-dejavu-serif.conf", {st_mode=S_IFREG|0644, st_size=858, ...},
0) = 0
newfstatat(AT_FDCWD, "/snap/firefox/6836/gnome-platform/etc/fonts/conf.d/20-unhint-small-vera.conf", {st_mode=S_IFREG|0644, st_size=1537, ...}, 0) = 0
newfstatat(AT_FDCWD, "/snap/firefox/6836/gnome-platform/etc/fonts/conf.d/25-arphic-ukai-render.conf", {st_mode=S_IFREG|0644, st_size=361, ...}, 0) = 0
newfstatat(AT_FDCWD, "/snap/firefox/6836/gnome-platform/etc/fonts/conf.d/25-arphic-uming-render.conf", {st_mode=S_IFREG|0644, st_size=362, ...}, 0) = 0
newfstatat(AT_FDCWD, "/snap/firefox/6836/gnome-platform/etc/fonts/conf.d/30-cjk-aliases.conf", {st_mode=S_IFREG|0644, st_size=16016, ...}, 0) = 0
newfstatat(AT_FDCWD, "/snap/firefox/6836/gnome-platform/etc/fonts/conf.d/30-metric-aliases.conf", {st_mode=S_IFREG|0644, st_size=13393, ...}, 0) = 0
newfstatat(AT_FDCWD, "/snap/firefox/6836/gnome-platform/etc/fonts/conf.d/35-arphic-ukai-aliases.conf", {st_mode=S_IFREG|0644, st_size=343, ...}, 0) = 0
newfstatat(AT_FDCWD, "/snap/firefox/6836/gnome-platform/etc/fonts/conf.d/35-arphic-uming-aliases.conf", {st_mode=S_IFREG|0644, st_size=353, ...}, 0) =
0
newfstatat(AT_FDCWD, "/snap/firefox/6836/gnome-platform/etc/fonts/conf.d/40-nonlatin.conf", {st_mode=S_IFREG|0644, st_size=5630, ...}, 0) = 0
newfstatat(AT_FDCWD, "/snap/firefox/6836/gnome-platform/etc/fonts/conf.d/41-arphic-ukai.conf", {st_mode=S_IFREG|0644, st_size=848, ...}, 0) = 0
newfstatat(AT_FDCWD, "/snap/firefox/6836/gnome-platform/etc/fonts/conf.d/41-arphic-uming.conf", {st_mode=S_IFREG|0644, st_size=1371, ...}, 0) = 0
newfstatat(AT_FDCWD, "/snap/firefox/6836/gnome-platform/etc/fonts/conf.d/45-generic.conf", {st_mode=S_IFREG|0644, st_size=3543, ...}, 0) = 0
```

## 9. Find process Using a port

## Command :

```
sudo fuser -n tcp 8080
```

**Explanation :**

sudo – Run the command with superuser privileges (necessary for seeing all processes and ports).

fuser – A command-line tool to identify processes using files or sockets.

-n tcp – Tells fuser to operate in the network namespace and specifically look at TCP ports.

8080 – The port number you're interested in (commonly used by HTTP servers or proxies).

**Output :**

```
ashishkumar@ashishkumar:~$ ps -fp 8080
UID          PID    PPID  C STIME TTY          TIME CMD
ashishkumar@ashishkumar:~$
```

## 10. Per-Process Statistics

## Command :

```
pidstat -p 3050 2 3
```

**Explanation :**

pidstat – Reports statistics for Linux tasks (processes), useful for analyzing CPU, memory, I/O usage, etc.

-p 3050 – Monitor only the process with PID 3050.

2 – Interval in seconds between reports.

3 – Number of reports (i.e., it will print 3 samples, 2 seconds apart).

**Output :**

```
ashishkumar@ashishkumar:~$ pidstat -p 3050 2 3
Linux 6.14.0-32-generic (ashishkumar)   09/26/2025      _x86_64_        (4 CPU)

09:21:58 PM   UID       PID    %usr %system  %guest   %wait    %CPU   CPU  Command
ashishkumar@ashishkumar:~$
```

### Explanation:

| Column | Meaning |
|---|---|
| `UID` | User ID who owns the process |
| `PID` | Process ID being monitored |
| `%usr` | CPU time spent in **user space** (non-kernel code) |
| `%system` | CPU time spent in **kernel space** (system calls, etc.) |
| `%guest` | CPU time used by guest OS (e.g. virtualized) |
| `%wait` | Time the CPU was idle **while the process was waiting for I/O** |
| `%CPU` | Total CPU usage by the process (`%usr` + `%system` + `%guest`) |
| `CPU` | The specific CPU core used |
| `Command` | Name of the process |

## 11. Control Groups (cgroups) for Resource Limits Create a new cgroup:

```
sudo cgcreate -g cpu,memory:/testgroup
```

**Explanation :**

| Part | Meaning |
|---|---|
| `sudo` | Run as root (required for cgroup operations) |
| `cgcreate` | Command to create a new cgroup |
| `-g cpu,memory:/testgroup` | Create the group under both the **CPU** and **Memory** controllers with the **name** **`testgroup`** |

## 🧠 12. Alternatives to nice / renice

### 1. chrt (Real-Time Scheduling)

```
sudo chrt -f 50 sleep 1000
chrt -p <pid>
```

**explanation :**

The command sudo chrt -f 50 sleep 1000 starts the sleep process with the FIFO (First-In-First-Out) scheduling policy and sets its priority to 50 (the range is usually 1 to 99, with higher being more urgent).

The command chrt -p <pid> can be used to view or change the real-time attributes of an already running process.

### 2.ionice (I/O Priority Control)

```
ionice -c 2 -n 7 tar -czf backup.tar.gz /home
```

**explanation :**

The ionice -c 2 -n 7 tar -czf backup.tar.gz /home command runs the backup process with a low disk I/O priority, ensuring it yields to more critical disk operations. It minimizes interference with other tasks by running the I/O in the "best-effort" class at the lowest priority level.

**output :**

```
ashishkumar@ashishkumar:~/UPES$ ionice -c 2 -n 7 tar -czf backup.tar.gz /home
tar: Removing leading `/' from member names
tar: /home/ashishkumar/.cache/ibus/dbus-2pki4VYl: socket ignored
tar: /home/ashishkumar/.cache/ibus/dbus-p3Rpf0sq: socket ignored
tar: /home/ashishkumar/.cache/ibus/dbus-3idLyHjO: socket ignored
tar: /home/ashishkumar/.cache/ibus/dbus-6nKIVDVc: socket ignored
tar: /home/ashishkumar/.cache/ibus/dbus-ix34a11m: socket ignored
tar: /home/ashishkumar/.cache/ibus/dbus-hl58iCgH: socket ignored
tar: /home/ashishkumar/.cache/ibus/dbus-n1qgbNin: socket ignored
tar: /home/ashishkumar/.cache/ibus/dbus-Bw5vVgI1: socket ignored
tar: /home/ashishkumar/.cache/ibus/dbus-61gs836I: socket ignored
tar: /home/ashishkumar/.cache/ibus/dbus-m2Ox04Oa: socket ignored
tar: /home/ashishkumar/.cache/ibus/dbus-GYFX13nH: socket ignored
tar: /home/ashishkumar/.cache/ibus/dbus-vAHw0tAh: socket ignored
tar: /home/ashishkumar/.cache/ibus/dbus-o9AMnemR: socket ignored
tar: /home/ashishkumar/.cache/ibus/dbus-dFXlrKIs: socket ignored
tar: /home/ashishkumar/.cache/ibus/dbus-qQTlU2sY: socket ignored
tar: /home/ashishkumar/.cache/ibus/dbus-myIejDGF: socket ignored
tar: /home/ashishkumar/.cache/ibus/dbus-UWylJi6p: socket ignored
tar: /home/ashishkumar/.cache/ibus/dbus-B34RLjVl: socket ignored
tar: /home/ashishkumar/.cache/ibus/dbus-C8xFunAd: socket ignored
tar: /home/ashishkumar/.cache/ibus/dbus-wYHut6au: socket ignored
tar: /home/ashishkumar/.cache/ibus/dbus-AuuTiIAN: socket ignored
tar: /home/ashishkumar/.cache/ibus/dbus-oFRPgKqA: socket ignored
tar: /home/ashishkumar/.cache/ibus/dbus-Lf9O0X7b: socket ignored
tar: /home/ashishkumar/.cache/ibus/dbus-Q4Cx2Nqi: socket ignored
tar: /home/ashishkumar/.cache/ibus/dbus-AGGaA7Uj: socket ignored
tar: /home/ashishkumar/.cache/ibus/dbus-oT5UtbLe: socket ignored
tar: /home/ashishkumar/.cache/ibus/dbus-6AIYjCTh: socket ignored
tar: /home/ashishkumar/.cache/ibus/dbus-UHkBLKil: socket ignored
tar: /home/ashishkumar/.cache/ibus/dbus-5ht16qoL: socket ignored
tar: /home/ashishkumar/.cache/ibus/dbus-lE9FqlJ5: socket ignored
tar: /home/ashishkumar/.cache/ibus/dbus-5W5JHcUj: socket ignored
tar: /home/ashishkumar/.cache/ibus/dbus-MnG0ciXD: socket ignored
tar: /home/ashishkumar/.cache/ibus/dbus-b0ujatyJ: socket ignored
tar: /home/ashishkumar/.cache/ibus/dbus-otPvHWho: socket ignored
tar: /home/ashishkumar/.cache/ibus/dbus-Konw1IJ5: socket ignored
tar: /home/ashishkumar/.cache/ibus/dbus-SoSpsmXA: socket ignored
tar: /home/ashishkumar/.cache/ibus/dbus-LiODwinP: socket ignored
tar: /home/ashishkumar/.cache/ibus/dbus-P9AXsRKg: socket ignored
tar: /home/ashishkumar/.cache/ibus/dbus-Mh6f0jkE: socket ignored
tar: /home/ashishkumar/.cache/ibus/dbus-EwWLPTSG: socket ignored
tar: /home/ashishkumar/.cache/ibus/dbus-gbUAFHUs: socket ignored
tar: /home/ashishkumar/.cache/ibus/dbus-D48me2qq: socket ignored
tar: /home/ashishkumar/.cache/ibus/dbus-Q0wh7IX1: socket ignored
tar: /home/ashishkumar/.cache/ibus/dbus-x5hvoEJ6: socket ignored
tar: /home/ashishkumar/.cache/ibus/dbus-6XJkKi55: socket ignored
tar: /home/ashishkumar/.cache/ibus/dbus-BEe6tYY1: socket ignored
tar: /home/ashishkumar/.cache/ibus/dbus-oX1SsK22: socket ignored
tar: /home/ashishkumar/.cache/ibus/dbus-vmcs9nvj: socket ignored
tar: /home/ashishkumar/.cache/ibus/dbus-doE3eu2F: socket ignored
tar: /home/ashishkumar/.cache/ibus/dbus-uIhcb52e: socket ignored
tar: /home/ashishkumar/.cache/ibus/dbus-2ToJN4Um: socket ignored
tar: /home/ashishkumar/.cache/ibus/dbus-sfTCFLUb: socket ignored
tar: /home/ashishkumar/.cache/ibus/dbus-xECjXHqP: socket ignored
tar: /home/ashishkumar/.cache/ibus/dbus-mrchbwIL: socket ignored
tar: /home/ashishkumar/.cache/ibus/dbus-ivwUyHB3: socket ignored
```

### taskset (CPU Affinity)

```
taskset -c 1 firefox
```

**explanation :**

The taskset command sets or retrieves the CPU affinity of a process, which means binding the process to specific CPU cores. This helps improve performance by limiting where a process can run, making better use of CPU caches and reducing overhead from switching cores.